The th Chinese Physics Olympiad Finals Problems

I. points

Fix a smooth vertical semicircular rail at , diameter ; is from vertical. A small smooth ring on it has mass . A light thin smooth elastic cord of stiffness passes through it, its ends tied at . Directly below , at , the cord is unstretched. Release it from rest at . At , it reaches ; makes angle with radius . Gravity is . Find the rail force magnitude for the cases .

Fig. Final

II. points

A small cart, mass , length , is on a level floor. Two springs are fixed near the bottom at both ends, collinear, with rest lengths , and stiffness , . The other ends meet small balls of masses , unattached. Initially ball compresses spring , holding the system at rest. Ball is locked to the cart floor.

Fig. Final

Ball is spring 's rest length from the right end. Release ball ; when spring regains rest length, unlock ball at once. Balls and collide and stick instantly. From release of ball to spring 's maximum compression, the cart moves . No friction. Find initial spring length and spring 's peak compression .

III. points

Station circles Earth at radius . Satellite circles in the same plane, with radius . and move counterclockwise. Launch a ship no station recoil to collect it. For fuel economy, except brief acceleration or braking, it coasts on identical anticlockwise transfer ellipses both ways. Both lie in the orbital plane. Perigee is on the station orbit, apogee on the satellite orbit. Given Earth radius and surface gravity , find:

Transfer orbit

Earth

Fig. Final

The departure increment from station to the ellipse. If at apogee it meets satellite , find capture speed increment for zero relative motion.

The least time to collect it and return, and the least angle station must turn around Earth.

IV. points